



Stop guessing, start shipping: Automated Pattern Discovery with Red Hat AI

Introducing AutoML & Auto RAG

William Caban

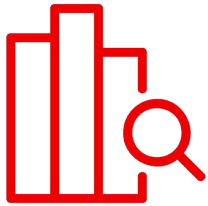
Sr Principal Product Manager,
Red Hat AI

Aditi Saluja

Product Manager,
Red Hat AI

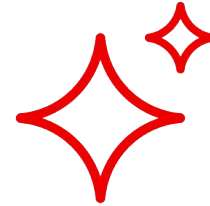
Enterprise AI Runs on Two Engines

Both are Hard to Build



Traditional AI

Approvals | Risk | Classification



Generative/Agentic AI

Q&A | Copilots | Agents

But, Hybrid AI Systems Are Hard to Scale Manually

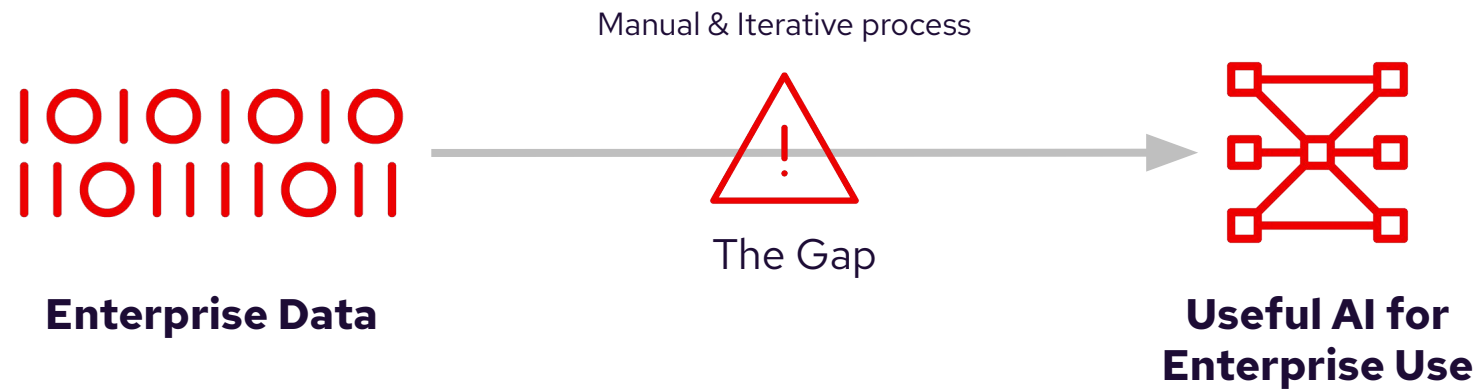
Traditional AI

- **Feature engineering** & selection is manual
- Model selection & **hyperparameter tuning** is iterative
- Performance degrades with **data drift**
- **Retraining pipelines** are slow and reactive
- Lab success $\neq$ production success

Generative/Agentic AI

- **Retrieval quality** varies across queries
- Chunking & embeddings require constant tuning
- **Hallucinations still occur** despite grounding
- **Prompt behavior** is inconsistent
- Teams **measure what is easy**, not what matters

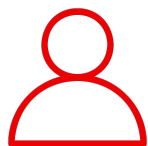
Enterprise AI Has a Gap. Experts Are the Only Bridge.



AI expertise is scarce, and that makes every gap bigger.

When experts are scarce, enterprises fall back on human evaluation

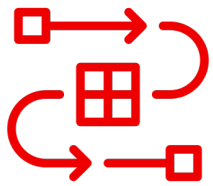
The responsible path – that doesn't scale



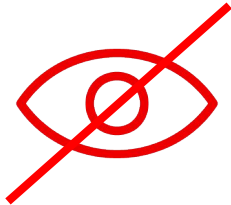
Evaluation Activity	Reality	Result
Domain experts manually review RAG outputs	Hours per query set, weeks per iteration	Too slow to keep pace with model changes
Multiple reviewers score predictions independently	High inter-reviewer disagreement ($\kappa < 0.6$)	Inconsistent signals, no reliable ground truth
Teams run spot-checks before deployment	5–10% of outputs sampled at best	False confidence production failures follow

Human evaluation is rigorous in theory.
At enterprise scale, it becomes the bottleneck, or gets skipped entirely.

The Risks of AI-Generated Code (“Vibe Everything”) Without Expertise



Agent Harnesses
AI Code Assistants



Without Expert
Oversight



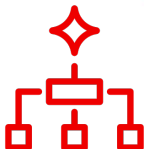
Without Rigorous
Evaluations



- Breaks in production
- Skips production hardening
- Hides problems behind vanity metrics

***The tools promised fast results. They delivered demos.
Enterprise production is a different problem.***

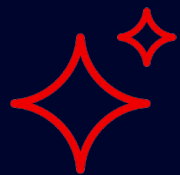
What if AI could encode expert knowledge – not just generate code?



Manual Expert	Vibe Coding	Auto Experiences
Deep expertise, rigorous process	Fast output, no embedded expertise	Expert knowledge encoded into automated and agentic systems
Doesn't scale, humans are the bottleneck	Scales fast, but delivers naive results	Scales with rigor: validated, ranked, production-ready
Weeks to a tuned, evaluated pipeline	Hours to a demo that breaks in production	Hours to ranked, evaluated, production-grade outcomes

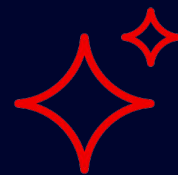
Auto Experiences are not a shortcut.
They work like an expert. Automated. Consistent. At scale.

AutoML and AutoRAG are the first.
The pattern scales.



AutoML ✓

Available in RHOAI 3.4



AutoRAG ✓

Available in RHOAI 3.4



AutoEval

Roadmap



Auto Risk Analysis

Roadmap



Auto Red-Teaming

Roadmap



Auto Judge Alignment

Roadmap

Today we go deep on the first two.



Red Hat

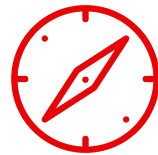
RAG Has Become the Default Pattern for Enterprise Generative AI Systems



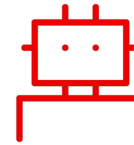
Internal
Copilots



Documentation
Assistants



Knowledge
Discovery



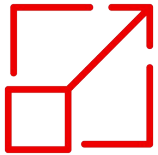
Customer
Support Chatbot



Search +
Summarization
workflows

However, RAG in the Enterprise Is Broken

What AI engineers and platform teams struggle with today



Weeks, Not Days

12-15 weeks to build a RAG pipeline from scratch



Too Many Choices

Chunking, embeddings, vector DBs - no guided path



Hallucination Risk

59% of enterprises cite hallucinations as top AI risk



No Evaluation

52% fail to continuously monitor RAG quality

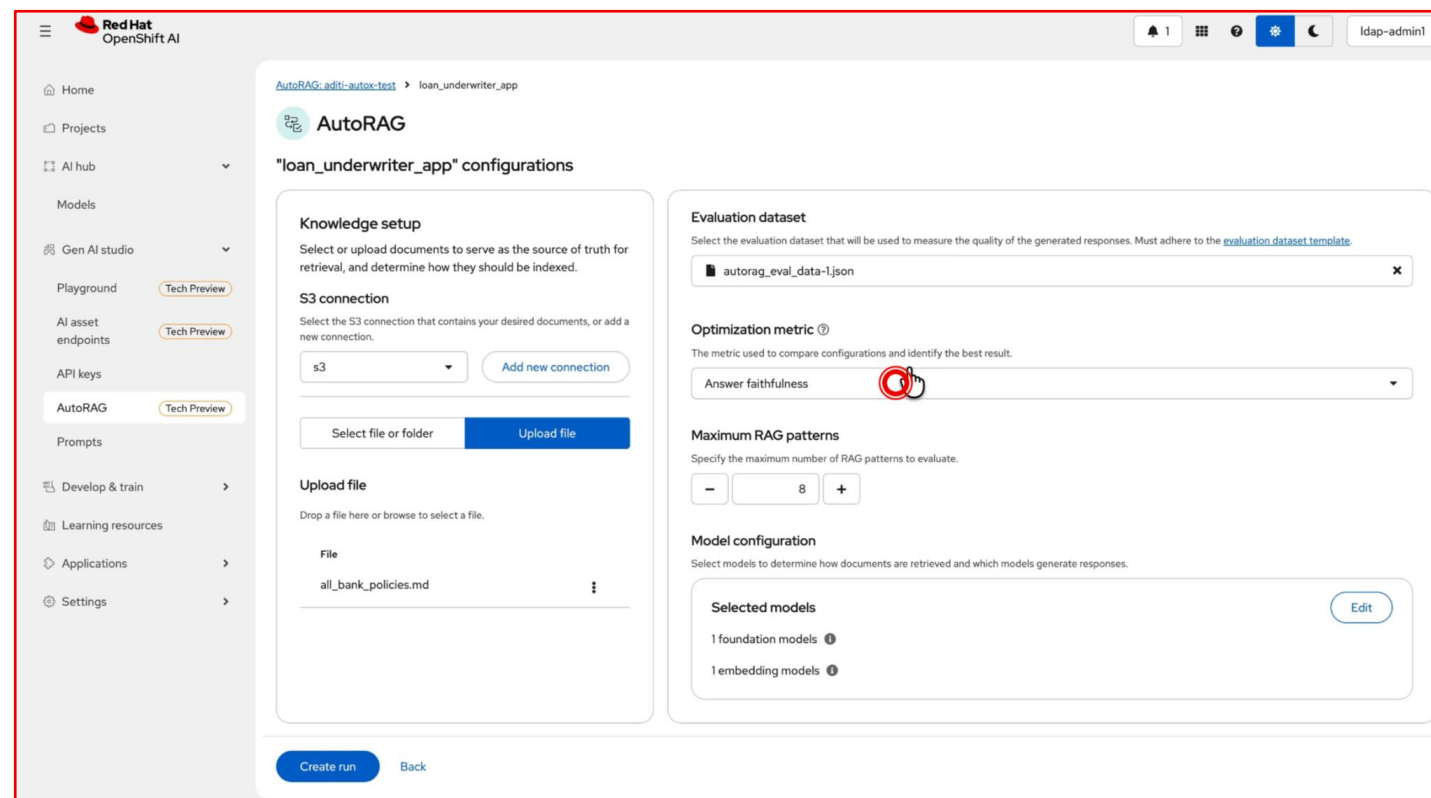
AutoRAG

What is it?

- ▶ **Automated Pipelines:** Automatically test multiple RAG algorithms and techniques for your use case
- ▶ **Hyper-parameter Optimization:** Evaluates variables such as chunk size, embedding models, vector stores and top-K retrieval settings.
- ▶ **Evaluation:** Uses advanced metrics (like Answer Correctness and Faithfulness)

Why it matters?

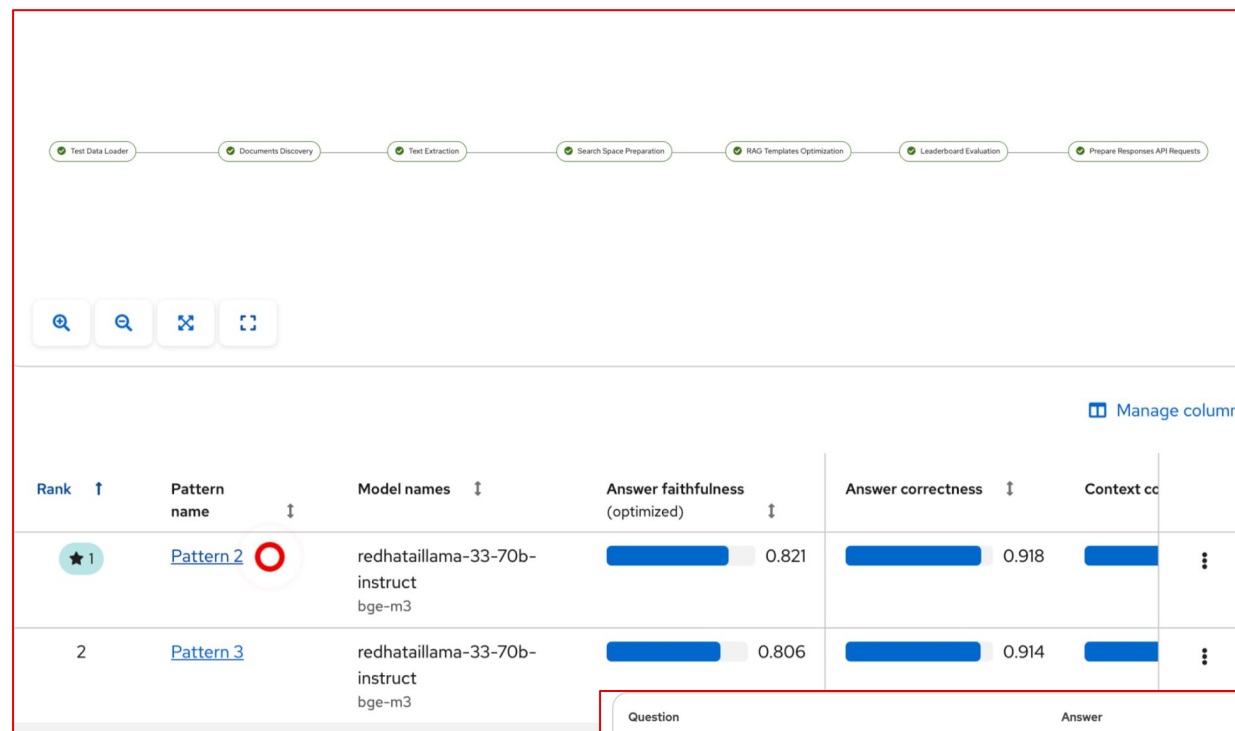
- ▶ Reduces complexity with RAG
- ▶ Accelerate Time-to-Market
- ▶ Ease of use



Auto RAG Key Features

What is it?

- ▶ **Search Space Prepration** - Chunking Size, Chunking Strategy, Embedding method, LLM Model
- ▶ **RAG Templates/Patterns Optimization:** Evaluates variables such as chunk size, embedding models, vector stores and top-K retrieval settings.
- ▶ **Ranked Leaderboard** of patterns tried based on chosen evaluation metric
- ▶ **Quick Results** on sample evaluation dataset provided by user



Question

What can I do to improve my chances of getting approved if I apply again in the future? I was rejected with credit score 665 and DTI 49.7%.

Answer

To improve your chances of getting approved, consider improving your credit score and reducing your debt-to-income (DTI) ratio. A credit score of 665 and DTI of 49.7% may be close to the thresholds, but the exact limits are not specified in the document. However, it is mentioned that DTI thresholds were updated in version 2 of the policy document. Reducing your DTI ratio and improving your credit score may help you qualify for approval in the future. Unfortunately, the exact thresholds are not provided, so it's unclear what specific targets to aim for.

[View expected answer \(1\)](#)

Expected answer 1

To improve your chances for future approval, here are specific steps you can take: (1) Reduce your DTI to below 36% - currently at 49.7%, you need to pay down approximately \$662 per month in debt to meet the Fair credit tier requirement per Policy UW-002. (2) Improve your credit score from 665 to at least 700 to qualify for the Good tier, which allows DTI up to 40% according to Policy UW-001. (3) As a self-employed applicant, ensure you have all required documentation ready for Policy UW-002. (4) Review your credit report for errors.

Answer correctness

Faithfulness

Context correctness

Auto RAG Roadmap

From Tech Preview to GA and beyond



RHOAI 3.4

May 2026 – Tech Preview

Backend engine + UI
experience launch



RHOAI 3.5

Jul 2026 – Harden

Quality, perf, endpoints,
multi-lingual



H2 2026

Expand & Mature

MLFlow, VectorDB
selection, GraphRAG



GA Readiness

Late 2026

Production guardrails and
GA certification

However, Traditional AI in the Enterprise Is Broken

What data science teams and ML engineers struggle with today



Weeks, Not Days

6–12 weeks from raw dataset to a validated, production-ready predictive model



Too Many Choices

100s of algorithms × 1000s of hyperparameter configurations – no guided path to the best model



Expert Dependency

Feature engineering and model selection require specialized ML expertise most enterprise teams lack



Guesswork, Not Science

Teams test a fraction of possible configurations and can't confirm they found the best performing model

Model development still a black art. Still a bottleneck. Enterprises deserve better.

Auto ML

What is it?

- ▶ Automate **end to end development** of machine learning workflows built using open source ML libraries
- ▶ **Feature Engineering**
- ▶ **Hyperparameter Optimization/Model Selection** - Selects best model for your use case by training several models with a range of hyperparameter configurations
- ▶ **Validation & Testing**: uses Metrics such as confusion matrix, ROC curve, F1 score

Why it matters?

- ▶ Reduces complexity of building ML workflows at scale
- ▶ Accelerate Time-to-Market
- ▶ Ease of use

RHOAI 3.4: UI and Notebooks for end-to-end AutoML pipelines (Tech Preview)

Auto ML Key Features (Input)

User Input - Dataset, Algorithm to run, Label to predict. Number of models to consider

Supported Algorithms

- ▶ Binary Classification
- ▶ Multiclass Classification
- ▶ Regression
- ▶ Time Series Modelling

The screenshot shows the AutoML configuration page for a project named "fraud-detection-may-4". It includes sections for "Documents" (with instructions to select or upload documents), "S3 connection" (with a dropdown menu showing "chris-cos-bucket" and an "Add new connection" button), and a "Select file from bucket" section (with a "Browse bucket" button). At the bottom, there is a table listing the selected file: "PS_20174392719_condensed_10..." of type "CSV".

Name	Type
PS_20174392719_condensed_10...	CSV

The "Configure details" section shows the "Prediction type" set to "Binary classification". It also shows the "Label column" set to "isFraud" and the "Top models to consider" set to 3. The "Binary classification" option is highlighted with a blue border.

Configure details

Prediction type *

- Binary classification**
Classify data into categories. Choose this if your prediction column contains two distinct categories.
- Multiclass classification
Classify data into categories. Choose this if your prediction column contains multiple distinct categories.
- Regression
Predict values from a continuous set of values. Choose this if your prediction column contains a large number of values.
- Time series forecasting
Predict future activity over a specified date/time range. Data must be structured and sequential.

Label column * ⓘ

isFraud

Top models to consider ⓘ

- 3 +

Auto ML Key Features (Output)

Leaderboard – Best model generated with Auto ML run

Feature Summary – Features ranked based on relative importance for model predictions

Model evaluation – F1 score, Confusion matrix

Rank	Model name	Accuracy (optimized)	Balanced Accuracy	F1	MCC	Precision	Recall
★ 1	LightGBM_BAG_L2_FULL	0.999	0.806	0.746	0.764	0.957	0.611
2	WeightedEnsemble_L3_FULL	0.999	0.806	0.746	0.764	0.957	0.611
3	RandomForestGini_BAG_L1_FULL	0.999	0.778	0.702	0.727	0.952	0.556

Model details

Rank1Accuracy (Optimized)0.999

DownloadSave as

MODEL VIEWER

Model information

Feature summary

EVALUATION

Model evaluation

Confusion matrix

Feature summary

Search feature names

Feature name	Importance
oldbalanceOrg	7.38%
newbalanceOrig	7.11%
type	2.49%
newbalanceDest	0.32%
amount	0.05%
step	0.04%
oldbalanceDest	0.01%
nameDest	0.00%

MODEL VIEWER

Model information

Feature summary

EVALUATION

Model evaluation

Confusion matrix

Confusion matrix

Observed	0	1	Percent correct
0	26639	14	99.9%
1	1	22	95.7%
Percent correct	100.0%	61.1%	99.9%

Less correctMore correct

Auto ML Roadmap



RHOAI 3.4

May 2026 – Tech Preview

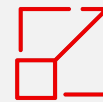
Backend engine + UI experience launch



RHOAI 3.5

Jul 2026 – Harden

Presets for algorithm, label selection, Out of the box support for Custom Runtime for Auto ML Models in RHOAI, Observability with MLFlow



H2 2026

Expand & Mature

Foundation models support (tabular and time series), Predictive Models as MCP tools for agents, Fairness metrics for regulated industries, Auto ML for multimodal



GA Readiness

Late 2026

Production guardrails and GA certification

Auto ML + Auto RAG Demo



<https://red.ht/AutoPatterns>



red.ht/uxfeedback



Participate in Red Hat AI
UX research opportunities!



Thank you

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